

Problema 4.1: a) Scrieti formula de interpolare Hermite pentru  $f \in C^4[-1, 1]$  si nodurile  $x_0 = -1$  simplu,  $x_1 = 0$  dublu si  $x_2 = 1$  simplu.

b) Determinati o formula de cuadratura de tip interpolat integrand formula precedenta termen cu termen.

c) Transformati formula precedenta intr-o formula pe  $[a, b]$ . Este aceasta o formula cunoscuta?

Rezolvare:

Tabela divergentelor divizate:

|            |         |                |                        |                                  |
|------------|---------|----------------|------------------------|----------------------------------|
| $z_0 = -1$ | $f(-1)$ | $f(0) - f(-1)$ | $f'(0) - f(0) + f(-1)$ | $\frac{f(1) - f(-1)}{2} + f'(0)$ |
| $z_1 = 0$  | $f(0)$  | $f'(0)$        | $f(1) - f(0) - f'(0)$  |                                  |
| $z_2 = 0$  | $f(0)$  | $f(1) - f(0)$  |                        |                                  |
| $z_3 = 1$  | $f(1)$  |                |                        |                                  |

Polinomul de interpolare va fi:

$$(H_3 f)(x) = f[z_0] + (x - z_0) \cdot f[z_0, z_1] + (x - z_0) \cdot (x - z_1) \cdot f[z_0, z_1, z_2] \\ + (x - z_0) \cdot (x - z_1) \cdot (x - z_2) \cdot f[z_0, z_1, z_2, z_3] =$$

$$f(-1) + (x+1)(f(0) - f(-1)) + x \cdot (x+1) \cdot (f'(0) - f(0) + f(-1)) + x^2 \cdot (x+1) \cdot \left(\frac{f(1) - f(-1)}{2} - f'(0)\right) \\ = \left(\frac{f(1) - f(-1)}{2} - f'(0)\right) \cdot x^3 + \left(\frac{f(1) + f(-1)}{2} - f(0)\right) \cdot x^2 + f'(0) \cdot x + f(0) \\ (R_3 f)(x) = \frac{(x+1) \cdot (x-1) \cdot x^2}{4!} f^{(4)}(\xi)$$

$$\text{b) } \int_{-1}^1 \left( \left( \frac{f(1) - f(-1)}{2} - f'(0) \right) \cdot x^3 + \left( \frac{f(1) + f(-1)}{2} - f(0) \right) \cdot x^2 + f'(0) \cdot x + f(0) + R \right) dx = \\ = \left( \frac{1}{8} \cdot (f(1) - f(-1) - 2 \cdot f'(0)) \cdot x^4 + \frac{1}{6} \cdot (f(1) + f(-1) - 2 \cdot f(0)) x^3 + \frac{1}{2} f'(0) \cdot x^2 + f(0) \cdot x + R \right) \Big|_{-1}^1 \\ = \frac{1}{3} \cdot (f(1) + f(-1) + 2 \cdot f(0)) + 2 \cdot f(0) = \frac{1}{3} \cdot f(-1) + \frac{8}{3} \cdot f(0) + \frac{1}{3} \cdot f(1) + R$$

$$\text{c) } \int_a^b \left( \left( \frac{f(1) - f(-1)}{2} - f'(0) \right) \cdot \left( \frac{2 \cdot t}{a-b} - \frac{a+b}{a-b} \right)^3 + \left( \frac{f(1) + f(-1)}{2} - f(0) \right) \cdot \left( \frac{2 \cdot t}{a-b} - \frac{a+b}{a-b} \right)^2 \right) \\ + \left( f'(0) \cdot \left( \frac{2 \cdot t}{a-b} - \frac{a+b}{a-b} \right) + f(0) + R \right) \frac{a-b}{2} dt = \frac{-(a-b)^2 \cdot (f(1) + f(-1) + 4 \cdot f(0))}{12}$$